

2018 Ph H2 Q4

Section: Our Dynamic Universe

Topic: Gravitation

A satellite orbits the Earth at a height of $3.6 \times 10^7 \text{ m}$ above the surface.
The mass of the Earth is $5.98 \times 10^{24} \text{ kg}$, and the radius of the Earth is $6.37 \times 10^6 \text{ m}$.

(a) Calculate the gravitational potential energy (GPE) of the satellite.

GPE relative to infinity is given by:

$$E_p = -\frac{GMm}{r}$$

Where:

- $G = 6.67 \times 10^{-11} \text{ N m}^2\text{kg}^{-2}$
- $M = 5.98 \times 10^{24} \text{ kg}$
- $m = 800 \text{ kg}$
- $r = R_{\text{Earth}} + h = 6.37 \times 10^6 + 3.6 \times 10^7 = 4.237 \times 10^7 \text{ m}$

$$E_p = -\frac{(6.67 \times 10^{-11}) \times (5.98 \times 10^{24}) \times 800}{4.237 \times 10^7}$$

$$E_p = -7.53 \times 10^9 \text{ J}$$

(b) Explain why this gravitational potential energy is negative

 Gravitational potential energy is **zero at infinity**, and becomes more negative as an object moves **closer to a mass**.
The energy is **negative** because it represents the **energy required to remove the satellite** from the gravitational field — you must do **positive work** to escape.

 Revision Tips

- Always use $r = \text{distance from centre of Earth} = R + h$
- GPE in space uses the **universal formula**:
$$E_p = -\frac{GMm}{r}$$
- GPE is negative because energy must be **supplied** to reach zero (escape Earth’s field)
- The more negative the value, the **stronger the gravitational bond**